

Unit Preview

In this Unit, you will discover

- what molecules are necessary for metabolic functions in cells,
- which major reactions occur in cells,
- how thermodynamic principles maintain metabolic function,
- which processes are involved in cellular respiration and photosynthesis, and
- how knowledge of metabolic processes can contribute to technological development.

Unit Contents

Chapter 1

The Chemistry of Life 4

Chapter 2

Enzymes and Energy 34

Chapter 3

Cellular Energy 62

Unit

Investigation 98

UNIT INVESTIGATION

Look ahead to pages 98–99.

- You can start planning your investigation well in advance by organizing what you will need.
- As you work through the unit, watch for ideas and materials that will help you prepare your experimental design.

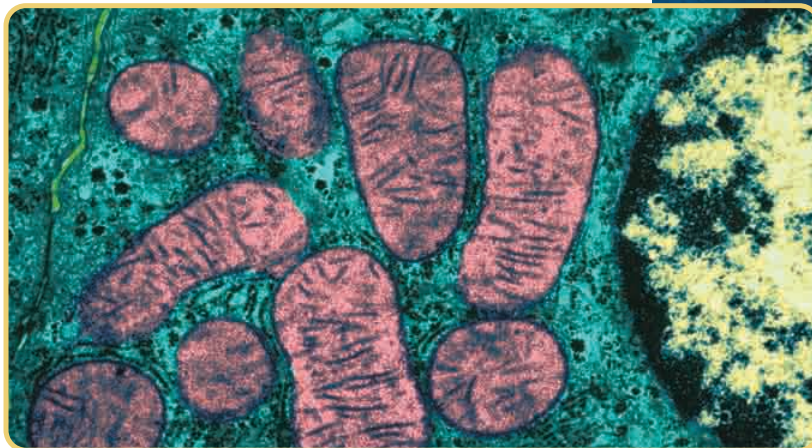
Metabolic Processes

Like large emeralds encrusted with gold, thousands of chrysalides (cocoons) hang from milkweed plants in southern Ontario. Within each of these chrysalides, a monarch butterfly caterpillar will undergo a metamorphosis to become an adult butterfly. This process requires much energy to fuel the tremendous changes that occur in a caterpillar's physical appearance and abilities.

All organisms require energy to survive. Cells in a eukaryotic organism contain organelles, such as the mitochondria shown below, that transform the energy in food into energy that can be used for various cellular processes. Without mitochondria, organisms such as the monarch caterpillar would not be able to perform the metabolic processes they need for metamorphosis. Metabolic processes involve all the chemical reactions that take place in cells, as well as the chemical reactions that need energy to transport molecules and build the cellular structures necessary for all life processes.

In this unit, you will learn about the chemical reactions that form molecules and see how the laws of thermodynamics govern all reactions between molecules. You will discover how special proteins are essential to metabolic processes in the cell. You will explore the series of metabolic reactions that take place in cells and learn how energy is transformed and used in these reactions. Finally, you will explore how the study of cell biology relates to your life and lifestyle.

How do organisms obtain the energy they need for life processes?





The Chemistry of Life

Reflecting Questions

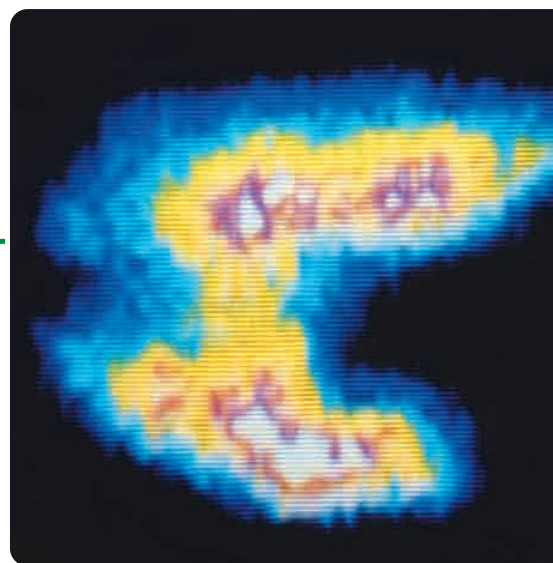
- How are bonds formed between atoms?
- What is ATP, and what reactions within the cell govern its formation?
- What reactions build and break down molecules?
- Why are large molecules essential for life?

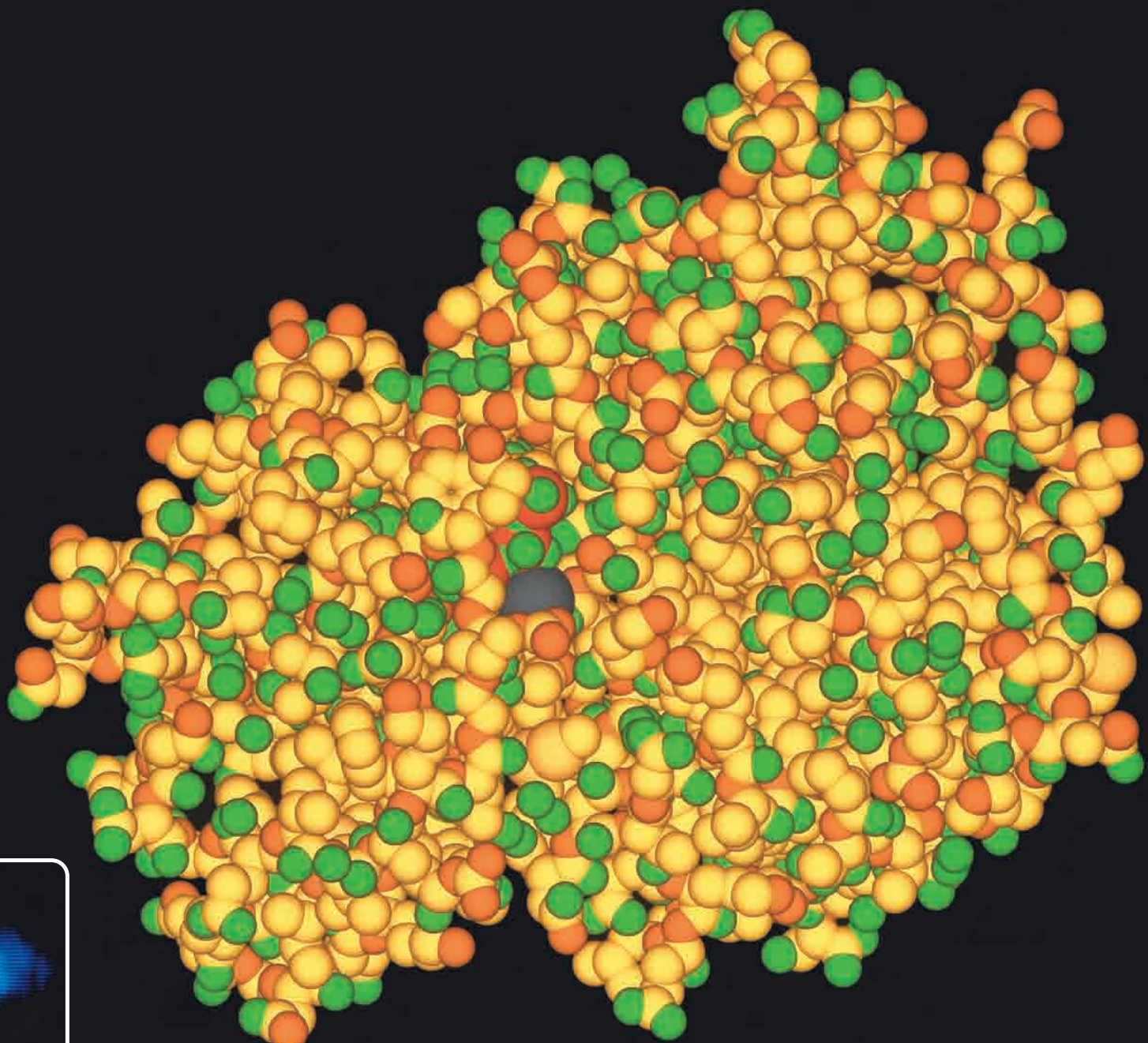
Within each living cell, chemical reactions take place millions of times every second. These reactions involve the transport of material, the removal of wastes, and the formation of structures, such as the molecule of protein shown in the image on the facing page. How do these reactions proceed and where are they carried out? Questions such as these have been at the centre of many investigations into metabolic processes in unicellular and multicellular organisms. Knowing how metabolic reactions take place and understanding the intricacies of the key steps in those reactions have helped researchers further their knowledge of metabolic processes.

In the field of human biology, for example, radiologists have been using radioactive isotopes since the 1930s to treat and diagnose disease. The electron micrograph on the right shows healthy human thyroid tissue. Radioactive iodine is used as a marker to locate thyroid tissue that may be cancerous. Iodine can only be utilized in thyroid tissue. If the machine tracing the radioactive iodine shows that some of it remains in the patient's body, doctors and technicians know that cancerous tissue remains to be removed. How are substances, like iodine, that are part of the non-living world related to substances that are part of the living world? What chemical processes are involved and what molecules play key roles in these processes?

In this chapter, you will explore the basis for biochemical processes. Although biologists are not usually concerned with individual atoms, they must know how atoms combine to form the molecules that animals and plant cells need to perform daily body functions such as growth, maintenance, and repair. Metabolic processes involve the interactions of electrons and the formation or breaking of bonds between atoms to produce or break down molecules. Understanding the chemistry of metabolism is key to understanding its many reactions and the products of those reactions.

How do biologists use their knowledge of molecules and chemical reactions in order to understand metabolic processes?





Chapter Contents

1.1 Atoms and Bonding

6

MiniLab:
Using Chemicals Safely

8

1.2 Biological Macromolecules and Their Subunits

14

MiniLab:
Reviewing Macromolecules

20

1.3 Making and Breaking Macromolecules

21

Investigation 1-A: Acids,
Bases, and Cellular pH

22

Investigation 1-B: Finding the
Products of Hydrolysis

24

Thinking Lab:
Polypeptides and Polarity

28

Atoms and Bonding

EXPECTATIONS

- Explain the term **electronegativity**.
- Express the relationship between **electronegativity, bonding, and the polarity of molecules**.

Living things are unique among all forms of matter. Unlike non-living things, all living things — from single-celled organisms such as the *Euglena* in Figure 1.1 to multicelled organisms such as whales and redwood trees — interact with and manipulate matter and energy. For example, all cells take in essential substances such as oxygen, water, and nutrients from their external environment. Inside cells, these substances undergo chemical reactions of several types. These reactions may be used to break down substances, synthesize others, and repair defective structures. Chemical reactions also provide energy for these life-sustaining activities, as well as others such as reproduction. Unneeded (or harmful) products of the reactions are eliminated as wastes.

Collectively, these processes — intake of substances, processing of substances, and elimination of wastes — are called metabolic processes, or **metabolism**. The substances involved in metabolism are molecules. The bonds that form between atoms define the structure and properties of these molecules. In this section, you will review several key ideas about atoms and bonding.



Figure 1.1 *Euglena*, a unicellular freshwater organism, carries out the same metabolic processes that your cells do.

Atoms and Elements

As you have learned in previous studies, all matter is formed of atoms. The **atom** is the smallest unit of matter involved in chemical reactions. Although tiny, atoms are complex structures, composed of even smaller subatomic particles. Most students of chemistry still study the model of the atom that Danish physicist Niels Bohr presented in the early twentieth century (see Figure 1.2). In this model, an atom consists of a small, dense core called a **nucleus**. It is composed of two kinds of subatomic particles — the positively (+) charged **protons** and the uncharged, or neutral, **neutrons**. Also in the Bohr model, negatively (–) charged **electrons** orbit the nucleus in one or more energy levels, or shells.

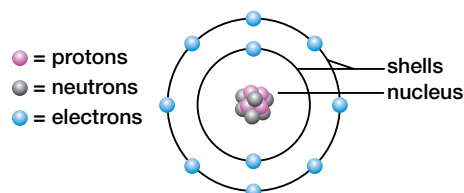


Figure 1.2 Niels Bohr's model of the neon atom

An **element** is a substance that cannot be broken down into simpler substances by chemical means. Substances such as calcium, oxygen, potassium, iron, and carbon are all elements. A few elements, such as helium, occur as single atoms. Several elements, such as hydrogen, nitrogen, and oxygen, occur as molecules made up of two atoms. Such molecules are called diatomic. Other elements such as phosphorus and sulfur occur as molecules made up of more than two atoms.

All atoms of an element have the same number of protons in their nuclei. This number, called the atomic number, is different for every element. The nuclei of carbon atoms, for example, each contain six protons. Because the nuclei of most atoms also contain neutrons, another important characteristic of an atom is its mass number. The mass number of an atom is the total number of protons and

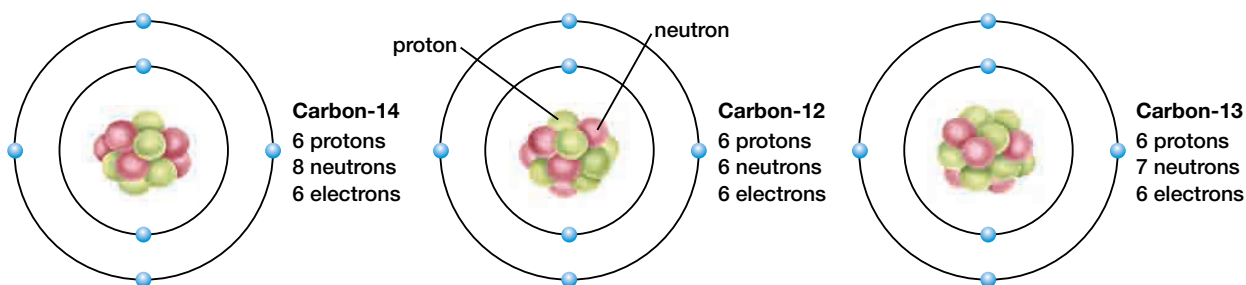


Figure 1.3 Carbon, one of the most important elements in living matter, has three naturally occurring isotopes. The nucleus of each isotope contains 6 protons, but the number

of neutrons in the nucleus is 6, 7, or 8. In each isotope, 6 electrons exist outside the nucleus.

neutrons in its nucleus. Atoms of the same element that contain different numbers of neutrons are called **isotopes** of that element. Refer to Figure 1.3 to see the numbers of protons, neutrons, and electrons in three isotopes of carbon. Their names include the mass number of each isotope: carbon-12, carbon-13, and carbon-14.

Some isotopes are stable, whereas others are unstable and break down (decay). The unstable isotopes are known as radioactive isotopes. Carbon-12 and carbon-13 are both stable isotopes, whereas carbon-14 is unstable and decays. Many radioactive isotopes decay at known rates. The rate at which a radioactive isotope decays may be used scientifically. The decay of carbon-14 can be used by archeologists, in a process called radiocarbon dating, to find the ages of some objects up to about 50 000 years old.

Table 1.1
The most abundant elements in living organisms

Element	Symbol	Atomic mass (u)	Abundance (% by mass)
oxygen	O	16.0	62.0
carbon	C	12.0	20.0
hydrogen	H	1.0	10.0
nitrogen	N	14.0	3.3
calcium	Ca	40.1	2.5
phosphorus	P	31.0	1.0
sulfur	S	32.1	0.25
potassium	K	39.1	0.25
chlorine	Cl	35.5	0.2
sodium	Na	23.0	0.1
magnesium	Mg	24.3	0.07
iron	Fe	55.9	0.01

Table 1.1 shows the atomic masses of the elements that are most abundant in living organisms. Notice that, unlike atomic numbers and mass numbers, some atomic masses are not whole numbers. This is the case because the atomic mass of an element is the average mass of all the naturally occurring isotopes of that element. Chlorine, for example, naturally occurs as a mixture of two isotopes: chlorine-35 and chlorine-37. There are three chlorine-35 atoms for every chlorine-37 atom. Therefore, the average mass of chlorine atoms is closer to 35 than to 37. The atomic mass of chlorine is, in fact, 35.5 u (atomic mass units). Appendix 7 provides atomic masses for all the known elements.

Electron Energy

Biologists usually study the groups of atoms that make up molecules rather than atoms and subatomic particles themselves. All cells obtain the energy to function from chemical reactions that involve molecules. The actions of electrons are key to this process.

According to the Bohr model, electrons orbit the nucleus of an atom within energy levels, or shells. An electron in the first shell (nearest the nucleus) has the lowest amount of potential energy. Any electrons in the remaining shells have more potential energy. Each shell can hold a maximum number of electrons. The first shell, for example, can hold a maximum of two electrons, while the second shell can hold a maximum of eight. Refer to Figure 1.2, which shows that in a neon atom the first two shells are filled. In general, the maximum number of electrons that a shell can hold is given by the formula $2n^2$, where n is the number of the shell. For example, the third shell can hold a total of $2(3)^2 = 18$ electrons.

The chemical properties of atoms rely mostly on the number of electrons in the outermost, occupied shell of an atom in its lowest energy state. This shell is known as the **valence shell**. The electrons that occupy the valence shell of any atom are called **valence electrons**. The elements in the periodic table that are least reactive are the noble gases, such as neon, found in group 18(8A) (see Appendix 7). Atoms of the other elements in the periodic table are more reactive than the noble gases. These elements can form chemical bonds with each other. The MiniLab examines safety issues involving the use of chemicals and how they react with each other during chemical bonding.

Ionic and Covalent Bonds

Most atoms can form **chemical bonds** with other atoms. These bonds are the forces that hold the atoms together in the form of compounds. For example, two chlorine atoms can combine (chemically react) to form a diatomic molecule of the element chlorine (Cl_2). Atoms of sodium and chlorine can combine to form the ionic compound sodium chloride (NaCl).

There are two general types of chemical bonds. One type involves the sharing of electrons between atoms, and is known as a **covalent bond**. The other type involves the transferring of one or more electrons from one atom to another, and is called an **ionic bond**. How are these bonds formed?

MINI LAB

Using Chemicals Safely

Throughout this course, you will use solutions of chemicals such as hydrochloric acid and sodium hydroxide to help you isolate, identify, or investigate the properties of various substances. Understanding how to use these chemicals properly, and what to do in case of an emergency, is vital to ensure your safety as well as the safety of your classmates.

Obtain a copy of an MSDS (Materials Data Safety Sheet) for the following:

- hydrochloric acid
- sodium hydroxide
- Benedict's solution
- Biuret solution (reagent)
- Lugol's solution
- silver nitrate

Copy the chart below into your notebook, and use the MSDS information to complete it.

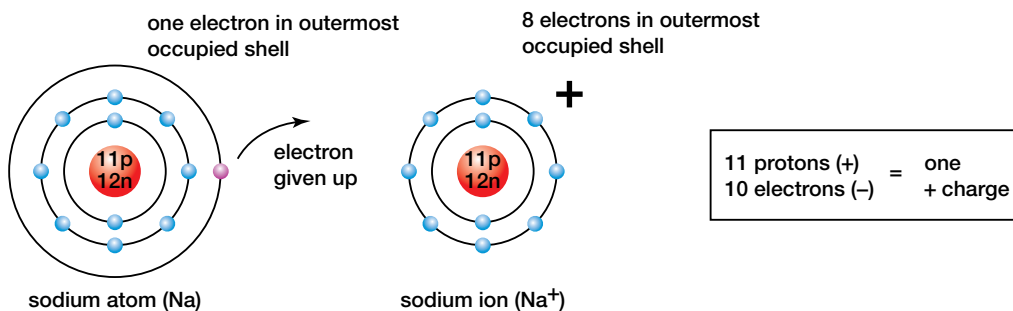
Analyze

- When you use these chemicals for a laboratory activity, what personal safety supplies will you need? Explain your choices.
- What concentration of substance are you likely to use in your laboratory activities?
- (a) Substance A is a clear, colourless liquid. How could you safely determine the identity of this liquid and/or its components?

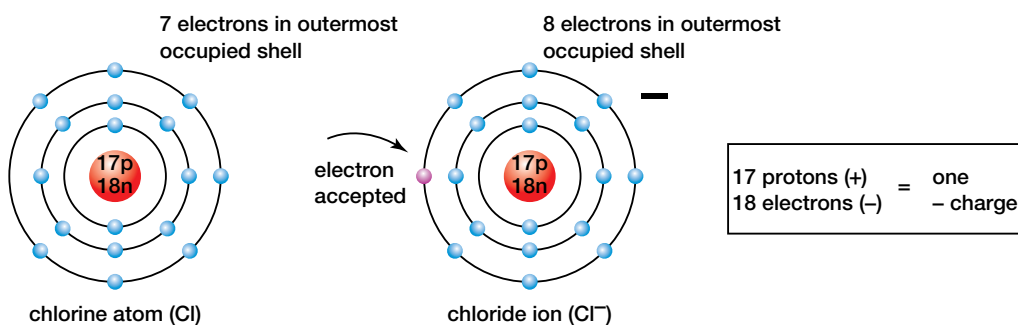
(b) Substance B is a blue liquid. How could you safely determine the identity of this liquid and/or its components?
- Write a short scenario that involves a materials spill for dilute hydrochloric acid. Explain how to respond safely to the spill.

	Hydrochloric acid	Sodium hydroxide	Benedict's solution	Biuret's solution	Lugol's solution	Silver nitrate
Physical/Chemical Properties						
■ Appearance?						
■ Odour?						
■ Solubility in Water?						
■ Toxicity?						
Health Hazard Data						
■ List three health hazards.						
Safety						
■ Identify three safety precautions related to the health hazards listed above.						
■ Spill Procedure?						
■ Neutralizing Agent?						

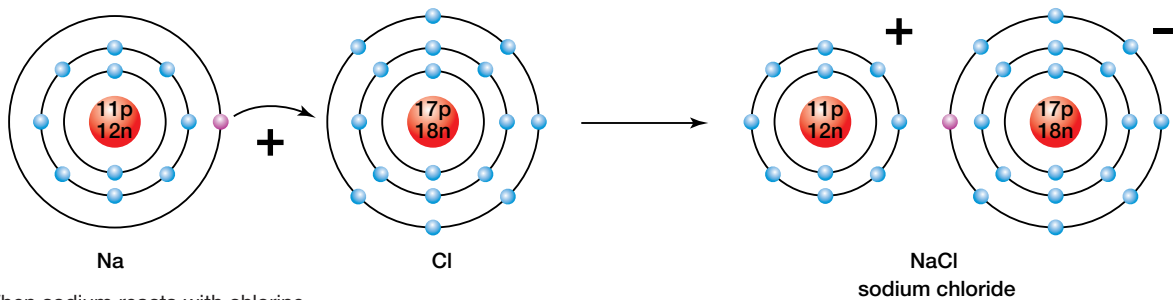
Ionic Bonding



- A** When a sodium atom gives up an electron, it becomes a positive ion.



- B** When a chlorine atom gains an electron, it becomes a negative ion.



- C** When sodium reacts with chlorine, sodium chloride (NaCl) is formed.

- D** In a sodium chloride crystal, ionic bonds between Na⁺ and Cl⁻ create a three-dimensional lattice. Each sodium ion is surrounded by six chloride ions, and each chloride ion is surrounded by six sodium ions.



1 mm
salt crystals

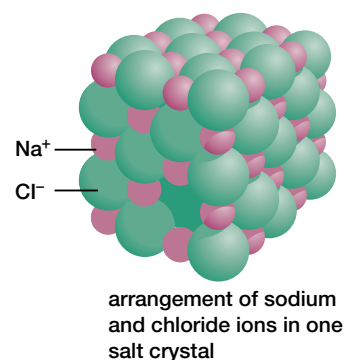


Figure 1.4 Reaction between sodium and chlorine to form ionic sodium chloride

Any atom has the same number of electrons and protons. Therefore, the atom has no charge and is said to be neutral. However, if an atom loses or gains electrons, that atom becomes an **ion**. If an atom loses electrons, the ion formed has more protons than electrons and therefore has a positive charge. A positively charged ion is called a **cation**. In contrast, if an atom gains one or more electrons the ion formed has a negative charge. A negatively charged ion is called an **anion**. When sodium (Na) and chlorine (Cl) atoms react, they form an ionic bond, as shown in Figure 1.4. The sodium atom gives up its only valence electron and becomes a sodium ion, with 11 protons and 10 electrons. This number of electrons is arranged in the same way that the 10 electrons are arranged in the neon atom. The chlorine atom gains an electron and becomes a

chloride ion, with 17 protons and 18 electrons. This number of electrons is arranged in the same way as the 18 electrons in an argon (Ar) atom. Because the sodium ion is positively charged and the chloride ion is negatively charged, they attract each other to form an ionic bond.

The tendency of chlorine to gain electrons is characteristic of atoms with a few electrons less than a noble gas atom. For example, atoms of fluorine and oxygen also tend to gain electrons when they form ionic bonds. One way to understand which elements form ionic bonds when they react is to use the principle of electronegativity.

Electronegativity is a measure of the relative abilities of bonding atoms to attract electrons. The Pauling scale is the most commonly used measure of electronegativities of atoms. Fluorine, the most

Covalent Bonding

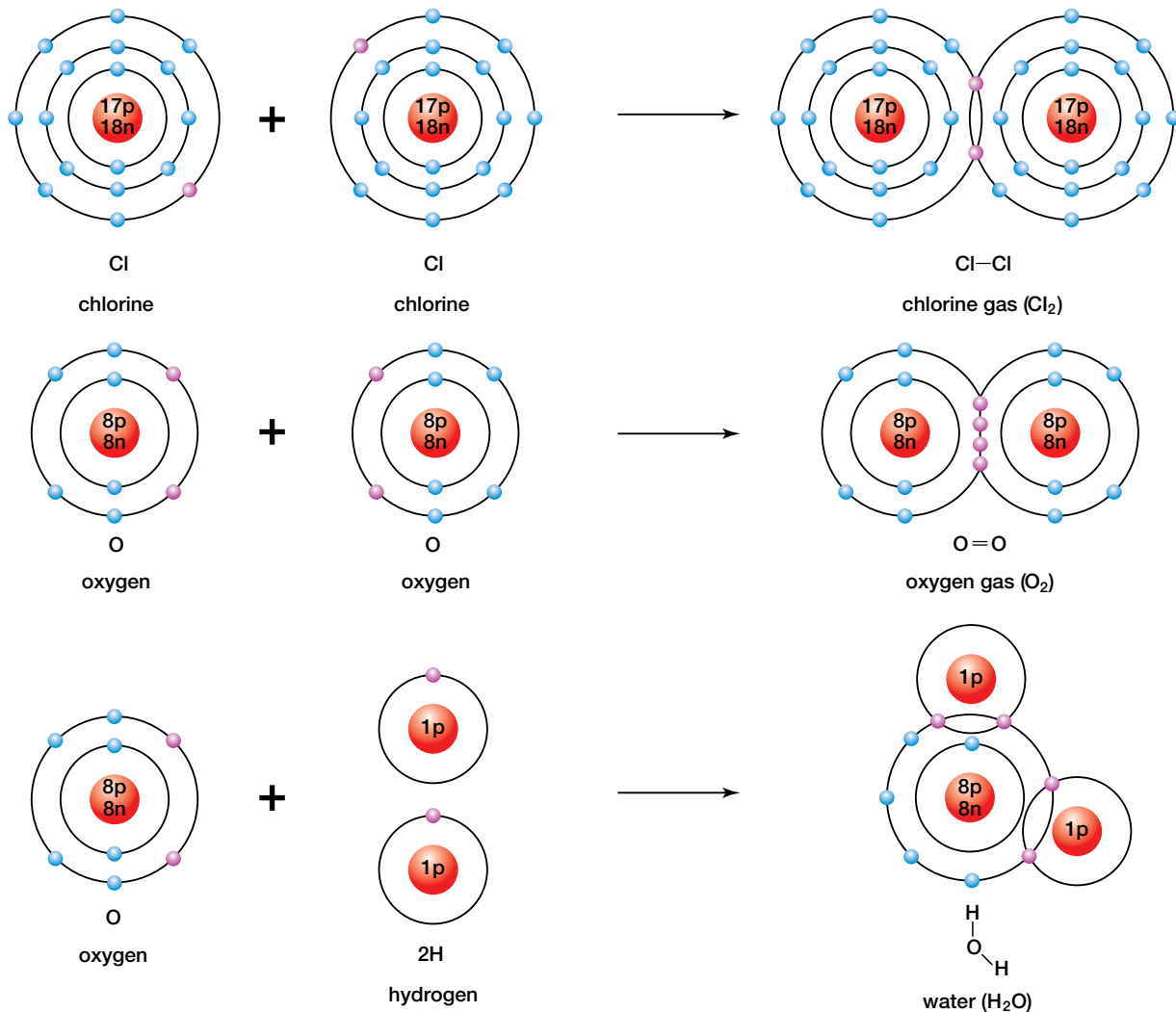


Figure 1.5 In a covalent bond, electrons are shared between atoms.